Frequency-and Edge-Guided Segment Anything Model for Remote Sensing Image Semantic Segmentation

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***Abstract*—Remote sensing image semantic segmentation (RSISS) has attracted significant attention due to the growing demand for fine-grained land-cover information. The segment anything model (SAM), proposed as a foundation vision model, offers strong segmentation performance and generalization capabilities for RSISS tasks. However, existing SAM-based approaches face two limitations: 1) insufficient adaptation of SAM's features to the diverse characteristics of land-cover types; and 2) semantic ambiguity at object boundaries, which hinders accurate delineation. To address these limitations, we propose the frequency- and edge-guided SAM (FE-SAM), a scalable and efficient framework for RSISS. In particular, we introduce a frequency-modulated adapter (FMA) that adaptively decomposes and modulates frequency-domain features based on the input data. It selectively enhances informative high- and low-frequency components corresponding to different land-cover types. Furthermore, to improve SAM's ability to capture fine-grained details, we design EGRefiner, which integrates multiscale edge-enhanced information extracted from the input image. Extensive experiments on three benchmark datasets demonstrate that FE-SAM outperforms state-of-the-art methods. The source code is available at: https://github.com/oucailab/FE-SAM**

***Index Terms*—Adapter tuning, fine-tuning, high-resolution remote sensing images, segment anything, semantic segmentation, visual foundation model.**

I. Introduction

REMOTE sensing image analysis is dedicated to processing and interpreting satellite or aerial images to offer an in-depth understanding of natural environments and human activities. As a pivotal technique, remote sensing image semantic segmentation (RSISS) facilitates diverse downstream applications, including natural disaster monitoring [1], land cover change detection [2], and urban planning [3]. Recent studies have explored the utilization of prior information and model-guided strategies to improve the representation capability of remote sensing images under challenging conditions [4], [5].

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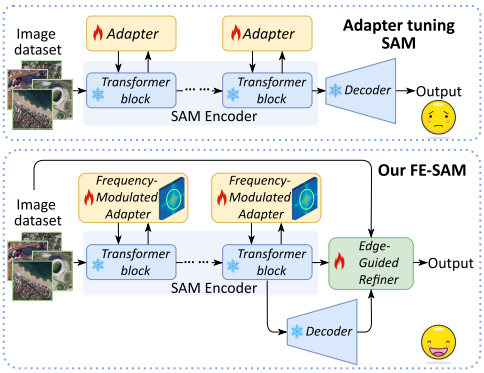


Fig. 1. Framework comparison between conventional adapter-tuned SAM and the proposed FE-SAM. Our FE-SAM reuses the pretrained weights of SAM with two novel modules: frequency-modulated adapter (FMA) and EGRefiner.

In recent years, many deep learning models have been proposed for the RSISS task. Most initial studies leverage the local modeling capacity of convolutional neural networks (CNNs) for segmenting ground objects [6], [7]. Subsequently, transformer-based approaches are put forward to capture long-range feature dependencies through the utilization of self-attention mechanisms [8], [9]. Although these methods have achieved excellent performance, there is still a gap from practical applications due to the limited network capacity and inadequate high-quality training data.

Recently, the segment anything model (SAM), developed for the purpose of image segmentation, has attracted considerable attention within the computer vision community [10]. It is trained on a massive dataset of over 11 million images and 1 billion segmentation masks, enabling it to recognize and segment objects across diverse categories. However, SAM is trained on natural images. As there are significant differences in distribution characteristics between the natural and remote sensing images, SAM can hardly be generalized to remote sensing data interpretation. Therefore, in this article, we aim to explore parameter-efficient fine-tuning (PEFT) of SAM for the RSISS task.

It is a nontrivial task for adapting SAM to the RSISS task due to the following two challenges.

1) Limitations in adapting SAM's features to diverse remote sensing land covers. SAM is pretrained on general natural images, and remote sensing objects exhibit remarkably different visual characteristics from SAM's training data. In addition, as shown in Fig. 1,

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different land-cover types exhibit distinct spatial distributions, leading to different proportions of high- and low-frequency components. Existing SAM-based remote sensing segmentation methods have limitations in adapting SAM's features to the remote sensing data from a frequency perspective.

2) Semantic ambiguity of ground object boundaries. Natural and man-made backgrounds in remote sensing scenes often contain textures similar to the ground objects of interest. For example, a road may share the same patterns as a building's roof, leading to boundary misclassification.

In addition, SAM's encoder employs a patch embedding strategy, which inevitably results in the loss of spatial detail information. These two challenges naturally correspond to the two key designs of our model. The first challenge requires adapting SAM features to the scene-dependent spectral characteristics of remote sensing imagery. The second challenge requires recovering fine spatial details degraded in the SAM pipeline.

To tackle these challenges, we propose frequency- and edge-guided SAM (FE-SAM) tailored for RSISS, which integrates frequency-modulated adapter (FMA) and edge-guided refiner (EGRefiner). In particular, FMA is inserted into the frozen SAM encoder to perform energy-guided frequency partitioning and prototype-based spectral recalibration, improving SAM feature adaptation to remote sensing scenes. Meanwhile, EGRefiner incorporates multiscale edge-enhanced cues from the input image to recover fine spatial details and refine object boundaries. In addition, to enhance fine-grained details of SAM, we propose EGRefiner to incorporate the multiscale edge-enhanced information from the input image. It can achieve more precise boundary segmentation of ground objects.

Compared with existing SAM adaptation and frequency-aware RSISS methods, FE-SAM focuses on integrating a frequency-aware adapter into the frozen SAM encoder and complementing it with an edge-guided refinement module for boundary recovery. In summary, the main contributions are as follows.

1) *Model Contribution:* We propose FMA as a lightweight adapter inserted into the frozen SAM encoder. It performs energy-guided frequency partitioning and prototype-based spectral recalibration, enabling parameter-efficient adaptation of SAM features to remote sensing scenes.
2) *Refinement Contribution:* We design EGRefiner to refine the object boundary. It incorporates multiscale edge-enhanced information from the input image and can achieve more precise boundary segmentation of desired objects.
3) *Experimental Contribution:* Extensive experimental validation on three benchmark datasets demonstrates the effectiveness and superiority of our model. We released the code to facilitate the remote sensing community.

II. Related Works

A. Segment Anything Model

The SAM [10], designed by Meta AI, is a pivotal advancement in the field of foundation models for computer vision, especially segmentation and object detection. It builds on the vision transformer [11] architecture and trains on 11 million images. SAM has shown remarkable capabilities in zero-shot transfer learning and boasts versatility across many vision tasks, including medical image analysis [12], [13], [14], camouflaged object detection [15], [16], [17], and remote sensing image interpretation [8], [18], [19]. To evaluate SAM's generalization ability in complex remote sensing scenes, RSPrompter [20] uses prompt learning for remote sensing instance segmentation via SAM. RingMo-SAM [21] constructs an instance- and terrain-type category-decoupling mask decoder for remote sensing object segmentation. However, these methods fail to enable SAM to adapt to diverse remote sensing land covers and show poor generalization in complex cases. Different from existing SAM-based methods, FE-SAM studies frequency-domain adaptation under a frozen SAM tuning setting and combines it with edge-guided refinement.

B. Remote Sensing Image Semantic Segmentation

Deep learning semantic segmentation of remote sensing images has attracted widespread attention. Fully convolutional networks (FCNs) [22], dilated convolution UNet [23], and attentive bilateral convolution network [24] have been employed for RSISS. Later, vision transformers have been used for remote sensing image analysis, and many hybrid CNN and transformer networks have been proposed for RSISS [8], [25], [26], [27]. Considering the class-dependent frequency characteristics of remote sensing scenes, as in recent frequency-aware studies [28], [29], this design enhances SAM's ability to adapt features to diverse land-cover categories. With the significant progress of large-scale foundation models in computer vision [30], [31], foundation models in the field of RS have the potential to solve this problem. RingMo [32], SatMAE++ [33], [34], and SpectralGPT [35] consider building foundation models for remote sensing images. However, existing SAM-based RSISS methods mainly focus on prompt design, adapter tuning, or decoder modification, while frequency-domain adaptation remains underexplored. Although lightweight transformer models have been studied for efficient multiscale feature interaction [36], they are not designed for SAM-based RSISS. Different from these methods and our previous frequency-aware framework, FE-SAM investigates frequency modulation under a frozen SAM adaptation setting and combines it with edge-guided refinement for land-cover representation and boundary delineation. In this article, leveraging SAM's powerful generalization capability, we can achieve effective semantic segmentation for the challenging RSISS task. In addition, we exploit multiscale edge-enhanced information to refine the object boundary, and the RSISS performance is improved.

C. Edge-Guided Segmentation Refinement

Accurate boundary delineation is important for dense prediction tasks, especially in remote sensing scenes where

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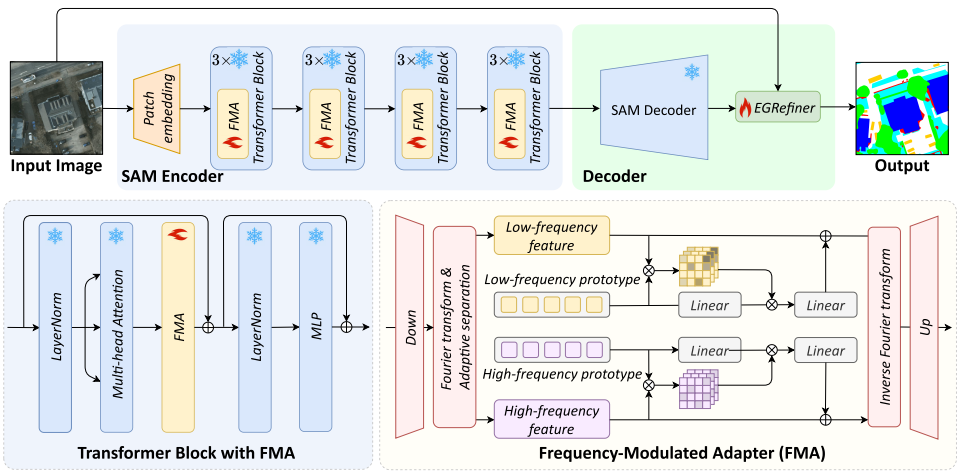


Fig. 2. Framework of our FE-SAM for RSISS. It reuses the pretrained weights of SAM with two novel modules: frequency-modulated adapter (FMA) and EGRefiner.

adjacent land-cover categories may exhibit similar textures or spectral responses. Edge-guided refinement has been widely used to enhance fine structural details by introducing explicit boundary cues into segmentation models. Recent studies also show that edge-guided or edge-semantics fusion strategies can improve object delineation in complex scenes [37].

In SAM-based RSISS, however, fine boundary details may be weakened by patch embedding and lightweight mask decoding. Therefore, besides frequency-domain adaptation, FE-SAM further introduces EGRefiner to extract multiscale edge-enhanced information from the input image and refine decoder predictions. This design complements FMA by improving spatial detail recovery and boundary quality.

III. Methodology

In this work, we propose FE-SAM for the RSISS task. Fig. 2 shows the overall framework of FE-SAM, which freezes the majority of parameters in the pretrained vision transformer encoder and introduces a small number of parameters through a frequency-modulated adapter (FMA). This parameter-efficient design avoids the high-computational cost associated with full-model fine-tuning while maintaining the strong representation capability of the pretrained SAM backbone. Recent studies on automated feature engineering have emphasized that effective feature design should balance representation capability and computational efficiency for practical machine learning systems [38]. Inspired by this perspective, FMA performs lightweight frequency-domain feature modulation to selectively adapt SAM features to diverse remote sensing scenes. This design preserves SAM's original prompt encoding mechanism and enables parameter-efficient adaptation to RSISS.

In addition, we present the EGRefiner to enhance the fine-grained details for better segmentation performance. FMA and EGRefiner are integrated in an encoder–decoder framework. FMA adapts encoder features from the frequency perspective, while EGRefiner introduces raw-image edge cues to refine decoder predictions. Their joint optimization enables complementary frequency adaptation and spatial detail recovery. We will detail the transformer block, FMA, and EGRefiner in Section III-A–III-C, respectively.

A. Transformer Blocks With FMA

The SAM encoder contains multiple sequential transformer blocks. Each transformer block consists of a multihead attention layer and an MLP layer. As shown in Fig. 2, the FMA follows the multihead attention and meets a skip connection. In the ith transformer block, the feature extraction in the multihead attention layer is computed as follows:

$$\mathbf{F'}_i = \mathrm{FMA}\left(\mathrm{Attn}\left(\mathrm{LN}\left(\mathbf{F}_i\right)\right)\right) + \mathbf{F}_i \tag{1}$$

where LN(·) denotes the layer normalization, Attn(·) denotes the self-attention, and FMA(·) represents the FMA. Afterward, the obtained features are fed into the MLP for nonlinear feature transformation as follows:

$$\mathbf{F}_{i+1} = \mathrm{MLP}\left(\mathrm{LN}\left(\mathbf{F'}_i\right)\right) + \mathbf{F}_i \tag{2}$$

here $\mathbf{F}_{i+1}$ is the output features of the ith transformer block and will be fed into the next transformer block.

With lightweight FMA, the proposed FE-SAM is capable of aligning SAM's features with the object appearances in remote sensing scenes and providing reasonable features for the subsequent decoder.

B. Frequency-Modulated Adapter

SAM is built upon large-scale natural image datasets. The natural images commonly have clear foreground–background contrasts and structured object layouts. However, remote sensing images generally have complex backgrounds and diverse land covers. In addition, different land-cover types exhibit distinct spatial distributions, leading to varying proportions of high- and low-frequency components. For instance, urban areas with complex structures (e.g., buildings and roads) generate dense high-frequency signals due to abrupt texture and edge changes. Rural regions predominantly contain low-frequency components from grasslands' smooth and large-scale patterns. Existing SAM-based remote sensing methods

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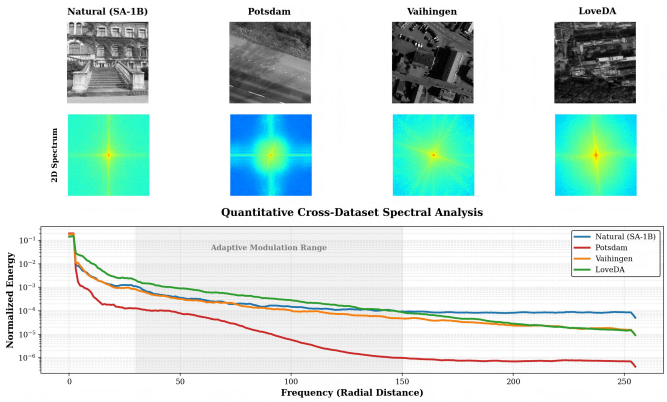


Fig. 3. Cross-dataset spectral analysis. (Top rows) Representative patches and their 2-D log-magnitude spectra from SA-1B and three remote sensing datasets. (Bottom) Averaged radial PSD curves.

mainly adapt SAM through prompts, lightweight adapters, or decoder modifications, but they rarely investigate how the spectral distribution shift between natural and remote sensing images affects SAM feature adaptation.

To solve the problem, we propose a frequency-modulated adapter (FMA) to adapt SAM to RSISS. FMA performs input-adaptive frequency partitioning within the frozen SAM encoder and recalibrates the separated spectral components through lightweight prototype-based modulation. This design aims to align SAM features with remote sensing scene statistics under a parameter-efficient tuning setting. As shown in Fig. 2, FMA consists of three parts: adaptive frequency separation, frequency feature adaptation, and reconstruction.

To analyze the spectral domain gap, we provide a cross-dataset spectral comparison in Fig. 3. We compute the averaged radial power spectral density (PSD) using 1000 randomly selected samples from SA-1B and three RS datasets with different resolutions and regions. As shown in Fig. 3, natural images exhibit more concentrated spectral responses, while RS images show more dispersed mid-to-high-frequency distributions. Despite slight spectral shifts caused by different ground sampling distances (GSDs) and sensors, the three RS datasets share a pattern distinct from the natural domain. This observation suggests a spectral distribution gap between natural and remote sensing images, especially in the mid- and high-frequency ranges. Rather than serving as direct causal evidence, Fig. 3 motivates the frequency-aware design of FMA, which adaptively determines the separation radius r and recalibrates frequency components for RS feature adaptation. Its effectiveness is further verified by the ablation studies in Section IV-D.

1) Adaptive Frequency Separation: The input data first undergoes a downsampling layer, and then the fast Fourier transform (FFT) is employed to convert the feature from the spatial domain to the frequency domain. Let $\mathcal{M}(u, v)$ denote the magnitude spectrum of the input data; the total energy is the sum of the magnitude-squared spectrum, and can be computed as

$$\mathbf{E}_t = \sum_{u=0}^{H-1} \sum_{v=0}^{W-1} |\mathcal{M}(u, v)|^2 \tag{3}$$

where H and W denote the height and width of the spectrum, respectively. To adaptively modulate the high- and low-frequency components of the input data, FMA determines the frequency separation boundary for each input feature map according to its spectral energy distribution. Although the energy threshold p is fixed as a global hyperparameter, the corresponding radius r is input dependent. In particular, we compute the total spectral energy E_t and determine the smallest radius whose accumulated energy from the low-frequency center reaches p

$$r = \min \left\{ \rho \;\middle|\; \frac{\sum_{(u,v)\in \mathrm{Disk}(\rho)} |\mathcal{M}(u, v)|^2}{E_t} \geq p \right\} \tag{4}$$

where $\mathrm{Disk}(\rho)$ denotes a circular region with radius ρ centered at the low-frequency origin of the shifted spectrum. The spectrum inside $\mathrm{Disk}(r)$ is regarded as the low-frequency component, while the remaining part is treated as the high-frequency component. In our experiments, p is set to 0.3, i.e., 30% of the accumulated spectral energy. Empirical validation shows that this setting provides a better balance between low-frequency structural preservation and high-frequency detail modeling in our experiments.

In $\mathcal{M}(u, v)$, the spectrum outside the radius r is extracted as high-frequency features, while the spectrum inside the radius r is extracted as low-frequency features. Therefore, we adaptively separate the high- and low-frequency components for remote sensing object feature modeling. The cutoff radius r in FMA is dynamically determined for each image according to the energy criterion, enabling input-adaptive frequency partitioning. This design allows FMA to adjust the separation boundary according to scene-specific spectral distributions, thereby better handling the diverse frequency characteristics of different land-cover types.

In the FMA module, we adopt FFT for frequency decomposition rather than discrete wavelet transform (DWT) or contourlet transform. FFT provides an efficient global spectral representation, supports adaptive frequency separation, and enables direct spatial reconstruction through inverse FFT. On the contrary, DWT mainly focuses on local multiscale analysis, while contourlet transform introduces higher computational complexity and often requires extra fusion operations. Therefore, FFT is better aligned with the lightweight and globally adaptive design of FE-SAM.

2) Frequency Feature Adaptation: Now we have low-frequency feature $\mathbf{T}_L$ and high-frequency feature $\mathbf{T}_H$. We introduce two sets of learnable prototypes to distill the key features from the high- and low-frequency components, respectively. For the high-frequency features, we first compute a similarity map $\mathbf{S}_H \in \mathbb{R}^{m\times n}$ between the high-frequency prototype $\mathbf{P}_H \in \mathbb{R}^{m\times c}$ and high-frequency features $\mathbf{T}_H \in \mathbb{R}^{c\times n}$ from the frozen SAM feature, where m denotes the number of frequency prototypes and c is the channel dimension. Here, the high-frequency prototype $\mathbf{P}_H$ is a learnable matrix. $\mathbf{S}_H$ is computed as

$$\mathbf{S}_H = \mathrm{Softmax}\left(\frac{\mathbf{M}_H \times \mathbf{T}_H}{\sqrt{c}}\right) \tag{5}$$

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where Softmax denotes the softmax activation function, and c represents the channel dimension of the high-frequency features T_H, which acts as a scaling factor to prevent the dot product from growing too large in magnitude.

In our implementation, m is set to 16 for both branches. Inspired by lightweight model design strategies in practical deployment scenarios [39], the prototype number is selected to balance spectral representation capability and computational efficiency. Increasing m may provide finer frequency pattern modeling but introduces additional parameters, while fewer prototypes may limit the representation of diverse land-cover characteristics. Therefore, $m = 16$ provides an effective trade-off between adaptation capability and model efficiency.

The prototypes are initialized once before training using offline K-means clustering on randomly sampled low- and high-frequency features after FFT-based adaptive separation. In particular, the prototype number m is also adopted as the number of clusters in K-means initialization. The resulting m centroids are used to initialize P_L and P_H. During end-to-end training, the prototypes are treated as learnable parameters and updated by standard backpropagation, without dynamic K-means during training or inference. The prototype dimension is kept consistent with the channel dimension of encoder frequency features for stable feature-prototype matching.

Next, we project the prototype $\mathbf{P}_H$ into the feature space of $\mathbf{T}_H$ via a linear layer, followed by elementwise multiplication with the similarity map $\mathbf{S}_H$. The product with $\mathbf{S}_H$ enables the prototype to better align to $\mathbf{T}_H$, where the high-frequency components from remote sensing data are accentuated.

Afterward, we project the obtained features by another linear layer and fuse them with $\mathbf{T}_H$ via elementwise summation. This process can be denoted as

$$\hat{\mathbf{T}}_H = W_2\left((W_1\mathbf{S}_H) \odot \mathbf{T}_H\right) + \mathbf{T}_H \tag{6}$$

where W_1 and W_2 denote the weights of the first and second linear layers, respectively. As shown in Fig. 2, the low-frequency features are computed in the same manner as those in the high-frequency branch.

3) Combination and Reconstruction: The outputs from the high- and low-frequency adaptation are combined. Finally, the combined features are converted to the spatial domain via an inverse FFT (IFFT), and then undergo an upsampling layer to generate the output.

C. Edge-Guided Refiner

While the FMA effectively adapts SAM's features from natural images to remote sensing scenarios, there are still some limitations. First, SAM's encoder employs a patch embedding strategy, which inevitably results in the loss of spatial details. Second, the upsampling approach used in SAM's decoder struggles to recover critical fine-grained information.

To address these issues, we propose the EGRefiner, as shown in Fig. 4. $\mathbf{F}_s$ denotes the feature from the SAM's mask decoder. We use a 3×3 convolution to extract local features of F_s, and then we progressively upsample the features to the input resolution via bilinear interpolations and 3×3 convolutions as follows:

$$\mathbf{F}'_s = \mathcal{C}_{3\times3}\left(\mathrm{UP}\left(\mathcal{C}_{3\times3}\left(\mathrm{UP}\left(\mathcal{C}_{3\times3}\left(\mathbf{F}_s\right)\right)\right)\right)\right) \tag{7}$$

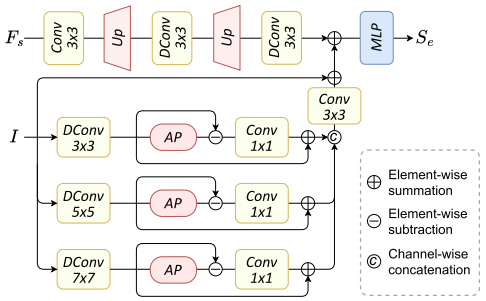


Fig. 4. Illustration of the EGRefiner.

where $\mathcal{C}_{3\times3}$ is the 3×3 convolutional layer, and UP denotes the $2\times$ upsampling with bilinear interpolation.

Before feature fusion, F_s is progressively upsampled to the input resolution, where each bilinear interpolation is followed by a 3×3 convolution for local spatial recalibration. Thus, the decoder feature F'_s and the edge-enhanced feature F'_{ee} have the same spatial resolution before concatenation. This implicit alignment mitigates feature mismatch and enables the fusion of high-level decoder semantics and low-level structural cues.

To inject spatial details into $\mathbf{F}'_s$, we extract multiscale fine-grained details from the input image, rather than from intermediate features produced by the SAM image encoder. This design is motivated by the loss of fine spatial details caused by the patch embedding and downsampling operations in SAM. Compared with intermediate encoder features, the original image retains more complete pixel-level structures for boundary refinement. Moreover, the multiscale depthwise convolutions enable EGRefiner to capture boundary details at different spatial scales, which is suitable for remote sensing scenes with objects of diverse sizes and shapes. In particular, given an input image I, we first employ a 3×3 convolutional layer to extract local features and obtain $\mathbf{F}_l$. Then, the edge information is extracted through three parallel depthwise convolutional paths of different scales (3×3, 5×5, and 7×7) as follows:

$$\mathbf{F}_e^1 = \mathcal{C}'_{3\times3}\left(\mathbf{F}_l\right), \quad \mathbf{F}_e^2 = \mathcal{C}'_{5\times5}\left(\mathbf{F}_l\right), \quad \mathbf{F}_e^3 = \mathcal{C}'_{7\times7}\left(\mathbf{F}_l\right) \tag{8}$$

where $C'_{3\times3}$, $C'_{5\times5}$, and $C'_{7\times7}$ denote depthwise convolution with different kernels, respectively. Next, we compute the edge-enhanced features by computing differences between the input and the pooled feature maps as follows:

$$\mathbf{F}_{ee}^i = \mathcal{C}_{1\times1}\left(\mathbf{F}_e^i - \mathrm{AP}\left(\mathbf{F}_e^i\right)\right) + \mathbf{F}_e^i, \quad (1 \le i \le 3) \tag{9}$$

where AP denotes the average pooling with 3×3 kernel. Afterward, we fuse these features with channelwise concatenation and employ a 3×3 convolutional layer as

$$\mathbf{F}'_{ee} = \mathcal{C}_{3\times3}\left(\mathrm{Concat}\left(\mathbf{F}_{ee}^1, \mathbf{F}_{ee}^2, \mathbf{F}_{ee}^3\right)\right) \tag{10}$$

where Concat$(\cdot)$ denotes channelwise concatenation. $\mathbf{F}'_{ee}$ contains both fine-grained details and the edge-enhanced information. We use $\mathbf{F}'_{ee}$ to complement the feature $\mathbf{F}'_s$ via elementwise summation. Finally, an MLP is employed to generate the final segmentation results $\mathbf{S}_e$. It is computed as follows:

$$\mathbf{S}_e = \mathrm{MLP}\left(\mathrm{Concat}\left(\mathbf{F}'_s, \mathbf{F}_l + \mathcal{C}_{3\times3}\left(\mathbf{F}'_{ee}\right)\right)\right). \tag{11}$$

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In (11), the integrated feature map in EGRefiner has 128 channels and is finally projected by a 1×1 convolutional layer to produce the probability maps for N target classes.

Multiscale edge enhancement is performed on the raw input image I rather than on frequency-modulated features. Thus, frequency-adapted semantic features are complemented by fine-grained structural cues from the original image. The final prediction S_e integrates decoder semantic features and EGRefiner-based edge details, enabling complementary frequency adaptation and boundary refinement for more accurate object delineation.

D. Loss Function

We employ a composite loss function that combines the binary cross-entropy (BCE) loss, the dice loss, and the L1 loss as follows:

$$\mathcal{L}\left(\mathbf{S}, \mathbf{S}_{gt}\right) = \mathcal{L}_{BCE} + \mathcal{L}_{Dice} + \mathcal{L}_{L1} \tag{12}$$

where $\mathbf{S}_{gt}$ is the ground truth of segmentation. To improve the learning capability, we use a 1×1 convolutional layer to refine the output of SAM's mask decoder as $\mathbf{S}_m = C_{1\times1}(\mathbf{F}_s)$. It should be noted that the auxiliary supervision is applied to the dense decoder feature map rather than discrete SAM masks. In particular, F_s is projected by a 1×1 convolution to obtain $S_m \in \mathbb{R}^{H\times W\times N}$, where N denotes the number of semantic classes, including the dataset-specific background or clutter category. Therefore, the auxiliary loss $\mathcal{L}(S_m, S_{gt})$ is computed over the entire image. This dense supervision provides gradients for both foreground and background regions, including areas that may not be well covered by the SAM masks, thereby improving the robustness of semantic prediction. Finally, we employ the loss on both the output of SAM's mask decoder and the output of EGRefiner as follows:

$$\mathcal{L}_{total} = \mathcal{L}\left(\mathbf{S}_e, \mathbf{S}_{gt}\right) + \mathcal{L}\left(\mathbf{S}_m, \mathbf{S}_{gt}\right). \tag{13}$$

IV. Experimental Results and Analysis

To ensure full reproducibility, we implemented FE-SAM using the PyTorch framework. The network was optimized using the AdamW optimizer with an initial learning rate of 1×10^{-4} and a weight decay of 0.01. The learning rate was dynamically adjusted using a cosine annealing schedule over a total of 100 epochs. All experiments were conducted on a single NVIDIA RTX 3090 GPU with a batch size of eight.

A. Datasets and Evaluation Metrics

1) ISPRS Vaihingen: The Vaihingen dataset consists of 33 very fine spatial resolution TOP image tiles, with an average size of 2494 × 2064 pixels. Each TOP image tile has three multispectral bands (near infrared, red, and green), as well as a digital surface model (DSM) and a normalized DSM (NDSM), with a GSD of 9 cm [40]. The dataset involves five foreground classes (impervious surfaces, buildings, low vegetation, trees, cars) and one background class (clutter). In our experiments, we only used the TOP image tiles and cropped the image tiles into patches of 1024 × 1024 pixels.

TABLE I
Pixel-Level Class Distribution on Three Datasets

	Class	Pixels($\times 10^6$)	Proportion(%)
Vaihingen	Imp. surf.	121.5	29.2
	Building	113.6	27.3
	Low veg.	79.1	19.0
	Tree	80.7	19.4
	Car	5.0	1.2
	Clutter	16.2	3.9
Potsdam	Imp. surf.	476.5	28.4
	Building	441.3	26.3
	Low veg.	322.1	19.2
	Tree	265.1	15.8
	Car	25.2	1.5
	Clutter	147.6	8.8
LoveDA	Background	1739.0	27.7
	Building	772.2	12.3
	Road	634.1	10.1
	Water	615.3	9.8
	Barren	339.0	5.4
	Forest	891.5	14.2
	Agriculture	1287.0	20.5

2) ISPRS Potsdam: The Potsdam dataset comprises 38 high-resolution TOP image tiles, with an average size of 6000 × 6000 pixels [41]. Each TOP image tile includes four spectral bands (red, green, blue, and near infrared), a DSM, and an NDSM, with a GSD of 5 cm. The dataset encompasses six primary classes (impervious surfaces, buildings, low vegetation, trees, cars, and clutter). In our analysis, we utilized the TOP image tiles and partitioned them into patches of 1024 × 1024 pixels.

3) LoveDA: The LoveDA dataset [42] contains 5987 fine-resolution optical remote sensing images (GSD 0.3 m) at a size of 1024 × 1024 pixels and includes seven landcover categories, i.e., building, road, water, barren, forest, agriculture, and background. The dataset encompasses two scenes (urban and rural) which are collected from three cities (Nanjing, Changzhou, and Wuhan) in China [43]. Therefore, considerable challenges are brought due to the multiscale objects, complex background, and inconsistent class distributions.

The pixel-level class distribution of the three datasets is presented in Table I. To objectively evaluate semantic segmentation performance, we adopt mean intersection over union (mIoU), overall accuracy (OA), and $F1$ score ($F1$) as the main evaluation metrics. These metrics provide quantitative comparisons across different methods and datasets, reducing the subjectivity of qualitative visualization. To further assess boundary segmentation quality, we introduce boundary IoU (B-IoU) [44] as a supplementary metric. Unlike standard mIoU, B-IoU evaluates pixels within a distance d from object boundaries. We set $d = 8$ and $d = 16$ pixels to measure the delineation accuracy of fine-grained structures, such as building corners and narrow roads.

B. Comparison With State-of-the-Art Methods

To comprehensively evaluate the effectiveness of our proposed FE-SAM method, we conducted extensive comparisons against a wide array of representative state-of-the-art models. These competitors span three primary categories.

1) CNN-based methods, including FCN8s [45], UNet [46], DeepLabv3 + [47], MA-Net [48], and A2-FPN [49].

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TABLE II

SEMANTIC SEGMENTATION RESULTS (MEASURED BY MIOU) OF VARIOUS METHODS ON THREE BENCHMARK DATASETS. THE BOLD AND UNDERLINE DENOTE THE BEST AND SECOND RESULTS

Method	Type	Vaihingen	Potsdam	LoveDA
FCN8s	CNN-based	78.25	81.03	48.02
U-Net		80.96	83.96	50.46
DeepLabv3+		81.78	84.02	51.45
MA-Net		82.76	84.91	51.64
A2-FPN		82.40	85.16	52.20
UNetFormer	Transformer-based	82.86	85.24	52.44
AFENet		84.27	86.43	<u>54.83</u>
CMTFNet		83.45	85.72	52.80
FGSANet		84.04	86.28	54.40
SFFNet		84.10	85.78	54.42
SparseFormer		83.83	85.87	54.17
SAM	SAM-based	83.04	85.94	52.54
EGSAM		83.69	86.12	54.10
SAM-Adapter		83.35	86.18	53.75
SAMed		83.17	86.21	54.39
RSPrompter		84.23	86.26	54.48
Semantic-SAM		<u>84.56</u>	<u>86.58</u>	54.59
FE-SAM (Ours)		**85.24**	**86.73**	**55.66**

2) Transformer-based segmentation methods, including UNetFormer [50], AFENet [29], CMTFNet [51], FGSANet [28], SFFNet [52], and SparseFormer [53].
3) Recent SAM-based methods, including the original SAM [10], EGSAM [37], SAM-adapter [54], SAMed [55], RSPrompter [20], and semantic-SAM [56].

1) Quantitative Evaluation: The overall semantic segmentation performance, measured by mIoU, is summarized in Table II. The results unequivocally demonstrate that recent SAM-based methods, including our FE-SAM, consistently outperform traditional CNN- and transformer-based approaches, primarily due to the highly generalized feature representation from SAM's large-scale pretraining. Our FE-SAM surpasses all other methods, achieving the best mIoU scores across all three benchmarks, which validates the overarching efficacy of our proposed frequency-domain adaptation and edge-guided refinement strategy. To provide a more granular, dataset-specific analysis, we present detailed per-class results.

On the ISPRS Vaihingen dataset, our FE-SAM's superiority is most pronounced on man-made objects requiring precise boundary delineation (see Table III). For instance, FE-SAM improves the $F1$-score for "building" to 96.03%, surpassing the next-best SAM-based competitor, and significantly outperforms on "car" with the score of 89.55%, a notable + 0.97% improvement over the runner-up. This is a direct consequence of our EGRefiner module, which is specifically engineered to counteract the spatial detail loss inherent in SAM's architecture by integrating multiscale edge information, thereby ensuring high-fidelity geometric reconstruction.

On the ISPRS Potsdam dataset, the per-class $F1$-scores show that our FE-SAM again achieves top-tier performance (see Table IV). While excelling on structured classes like "Building," its strength is also evident in resolving ambiguity between spectrally similar classes. Our model achieves the best $F1$-scores for "Low vegetation" and "Tree," outperforming strong competitors like SparseFormer. This highlights the critical contribution of our FMA module. By empowering the model to differentiate land-cover types based on their inherent frequency profiles, the FMA effectively mitigates the semantic confusion that plagues methods relying solely on spectral features.

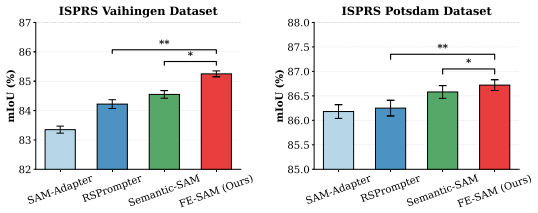


Fig. 5. Statistical significance analysis of mIoU on the ISPRS Vaihingen and Potsdam datasets.

On the LoveDA dataset, the per-class IoU results in Table V confirm the robust generalization of our FE-SAM, which obtains the highest mIoU of 55.66%. The efficacy of our approach is evident in its balanced performance. It achieves state-of-the-art IoU on large, continuous structures like "Road" and "Water," while simultaneously outperforming most competitors on dense objects like "Building." This balanced superiority is a direct testament to the FMA module's ability to selectively modulate frequency components—enhancing low-frequencies to preserve the coherence of large-scale structures and boosting high-frequencies to capture the details of small-scale objects.

To assess the stability of FE-SAM, we conducted ten independent trials for each method using random seeds. All trials followed the same data splits, training schedule, augmentation strategy, and evaluation protocol, while only the random seed was changed. For statistical testing, the results obtained under the same random seed were treated as paired observations, and a two-sided paired t-test was conducted on the mIoU differences between FE-SAM and the strongest competing methods. As shown in Fig. 5, the error bars denote the standard deviation of mIoU values across trials. In addition, we report the mean mIoU, standard deviation, 95% confidence interval, and exact p-value to provide a more complete statistical comparison. The results show that FE-SAM achieves consistent improvements over the competing methods across different random seeds. The statistical test further indicates that these improvements are unlikely to be caused solely by random seed variations under the current experimental setting. Though our overall metric gains over competing methods are moderate, such improvements carry practical value for remote sensing segmentation, as they boost land-cover mapping and boundary extraction. Consistent performance gains across datasets verify that our frequency adaptation and edge refinement effectively mitigate feature shifts and blurry boundaries.

2) Qualitative Evaluation: To complement the quantitative metrics, we provide a qualitative analysis to visually demonstrate the segmentation performance of FE-SAM. On the ISPRS Vaihingen dataset (Fig. 6), it shows the critical role of the EGRefiner module in ensuring high-fidelity geometric reconstruction. As highlighted by the bounded regions in the third row, existing SAM-based methods like SAM-Adapter and

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TABLE III

QUANTITATIVE COMPARISONS WITH THE STATE-OF-THE-ART METHODS ON THE ISPRS VAIHINGEN DATASET. THE BOLD AND UNDERLINE DENOTE THE BEST AND SECOND RESULTS

Method	Type	Backbone	Per-class F1-score					mIoU(%)	m-F1(%)	OA(%)
			Imp.surf.	Building	Low veg.	Tree	Car			
FCN8s	CNN-based	VGG-16	95.22	91.89	81.76	88.52	79.68	78.25	87.41	88.96
U-Net		ResNet-18	96.03	94.51	82.58	88.73	83.76	80.96	89.12	89.94
DeepLabv3+		ResNet-18	96.38	95.13	82.88	88.86	84.84	81.78	89.62	90.50
MA-Net		ResNet-18	96.39	95.17	83.86	89.51	86.33	82.76	90.25	90.69
A2-FPN		ResNet-18	96.31	95.04	83.01	89.08	86.67	82.40	90.02	90.52
UNetFormer	Transformer-based	ResNet-18	96.55	95.24	83.36	89.11	87.34	82.86	90.32	90.74
AFENet		ResNet-18	96.72	95.47	84.82	90.38	89.12	84.27	91.30	91.41
CMTFNet		ResNet-50	96.49	95.43	83.96	89.48	88.04	83.45	90.68	90.88
FGSANet		FGSABackbone-L	96.95	95.57	85.83	89.04	88.36	84.04	91.15	93.50
SFFNet		ConvNext-tiny	96.63	95.32	84.51	89.88	89.00	84.10	91.07	91.10
SparseFormer		Swin-tiny	96.46	95.39	84.16	89.70	88.58	83.83	90.86	90.96
SAM	SAM-based	ViT-B	96.83	95.24	85.17	89.53	85.84	83.04	90.52	93.14
EGSAM		ViT-B	96.78	95.15	85.47	89.77	87.53	83.69	90.94	93.40
SAM-Adapter		SAM-B	96.90	95.00	84.67	90.41	86.61	83.35	90.72	93.25
SAMed		SAM-B	96.77	95.21	84.75	90.35	85.93	83.17	90.60	93.19
RSPrompter		SAM-B	97.06	95.72	85.93	90.19	87.44	84.23	91.26	93.56
Semantic-SAM		SAM-B	97.04	95.81	85.66	90.22	88.58	84.56	91.46	93.57
FE-SAM(Ours)		SAM-B	**97.10**	**96.03**	**86.10**	**90.61**	**89.55**	**85.24**	**91.88**	**93.79**

TABLE IV

QUANTITATIVE COMPARISONS WITH THE STATE-OF-THE-ART METHODS ON THE ISPRS POTSDAM DATASET. THE BOLD AND UNDERLINE DENOTE THE BEST AND SECOND RESULTS

Method	Type	Backbone	Per-class F1-score					mIoU(%)	m-F1(%)	OA(%)
			Imp. surf.	Building	Low veg.	Tree	Car			
FCN8s	CNN-based	VGG-16	90.62	92.55	85.02	85.95	92.48	81.03	89.32	88.00
U-Net		ResNet-18	92.31	94.56	85.82	87.37	95.01	83.96	91.01	89.60
DeepLabv3+		ResNet-18	92.78	95.07	86.10	87.41	94.86	84.02	91.24	89.70
MA-Net		ResNet-18	92.86	95.30	86.47	87.77	95.45	84.91	91.57	90.41
A2-FPN		ResNet-18	93.08	95.79	86.53	88.00	94.84	85.16	91.65	90.44
UNetFormer	Transformer-based	ResNet-18	92.96	95.77	86.84	87.96	95.30	85.24	91.77	90.33
AFENet		ResNet-18	93.74	96.22	87.75	88.63	**96.22**	86.43	92.51	91.27
CMTFNet		ResNet-50	93.55	95.88	87.23	88.01	95.56	85.72	92.05	90.76
FGSANet		FGSABackbone-L	93.84	96.43	87.67	88.64	95.97	86.28	92.49	91.32
SFFNet		ConvNext-tiny	93.48	95.91	87.40	88.15	95.61	85.78	92.11	90.88
SparseFormer		Swin-tiny	93.43	95.95	87.30	88.24	95.90	85.87	92.16	90.84
SAM	SAM-based	ViT-B	93.71	96.28	87.55	88.31	95.73	85.94	90.32	91.01
EGSAM		ViT-B	93.83	96.34	87.62	88.46	95.85	86.12	92.42	91.25
SAM-Adapter		SAM-B	93.85	96.35	87.66	88.53	95.87	86.18	92.45	91.21
SAMed		SAM-B	94.00	96.42	87.55	88.49	95.88	86.21	92.47	91.28
RSPrompter		SAM-B	94.02	96.44	87.77	88.45	95.82	86.26	92.50	91.41
Semantic-SAM		SAM-B	94.01	96.46	87.42	88.39	96.11	86.58	92.48	91.37
FE-SAM(Ours)		SAM-B	**94.03**	**96.48**	**88.31**	**88.93**	96.15	**86.73**	**92.78**	**91.56**

SAMed struggle with boundary precision, producing masks with notably blurred edges and rounded corners that fail to adhere to the true object geometry. The sharp, right-angled corners of the building are incorrectly rendered as smooth curves. On the contrary, our FE-SAM renders these boundaries with exceptional sharpness and geometric precision, accurately capturing the crisp footprint of the building in a manner that closely mirrors the ground truth. It confirms that by integrating multiscale edge-enhanced information, the EGRefiner directly counteracts the loss of spatial detail inherent in SAM's architecture.

On the ISPRS Potsdam dataset (see Fig. 7), the visualizations expose the limitations of other methods in handling areas with low textural contrast, where severe semantic confusion is a prevalent failure mode. As illustrated in the highlighted region of the second row, a critical failure is the misclassification of large swathes of "low vegetation" as "impervious surface." Many methods, from classic U-Net to advanced SAM-based adapters, produce an almost entirely uniform and incorrect prediction in this area, failing to capture the presence of the vegetation. Furthermore, the third row highlights the common struggle to preserve discrete "cars" amidst background noise, where models like U-Net and A2-FPN incorrectly merge the vehicles into the "clutter" class. Our FE-SAM, however, demonstrates exceptional robustness against both types of errors. It correctly identifies the "low vegetation" in the second row and delivers a much cleaner segmentation of the individual vehicles in the third row. This comprehensive improvement stems directly from the FMA module. By analyzing and adaptively modulating the distinct frequency profiles, it successfully resolves the spectral ambiguity between low-texture classes like vegetation and impervious surfaces.

On the LoveDA dataset (see Fig. 8), it highlights FE-SAM's superior generalization. In the second and third rows, competing approaches like A2-FPN and the base SAM produce

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TABLE V

QUANTITATIVE COMPARISONS WITH THE STATE-OF-THE-ART METHODS ON THE LOVEDA DATASET. THE BOLD AND UNDERLINE DENOTE THE BEST AND SECOND RESULTS

Method	Type	Backbone	Per-class IoU							mIoU(%)
			Background	Building	Road	Water	Barren	Forest	Agriculture	
FCN8s	CNN-based	VGG-16	54.46	63.95	56.88	62.66	11.83	32.30	47.03	48.02
U-Net		ResNet-18	54.16	58.56	57.19	57.93	27.31	41.54	56.53	50.46
DeepLabv3+		ResNet-18	54.39	60.15	58.09	63.30	26.52	41.94	55.38	51.45
MA-Net		ResNet-18	54.76	60.94	57.10	62.18	34.82	37.26	54.42	51.64
A2-FPN		ResNet-18	55.64	62.00	58.42	65.56	28.08	42.19	53.54	52.20
UNetFormer	Transformer-based	ResNet-18	55.73	59.68	56.81	66.36	30.36	41.36	56.75	52.44
AFENet		ResNet-18	47.43	59.22	**59.17**	**81.55**	21.36	**48.68**	**66.40**	54.83
CMTFNet		ResNet-50	55.95	60.46	58.16	68.66	29.48	41.76	55.10	52.80
FGSANet		FGSABackbone-L	55.90	64.20	57.80	67.80	35.50	43.80	55.80	54.40
SFFNet		ConvNext-tiny	56.44	65.00	58.45	66.33	32.97	44.05	57.71	54.42
SparseFormer		Swin-tiny	56.59	63.03	56.73	66.32	32.30	43.85	60.36	54.17
SAM	SAM-based	ViT-B	53.66	59.41	57.91	66.91	32.50	39.74	57.66	52.54
EGSAM		ViT-B	55.80	63.80	57.50	67.50	35.00	43.50	55.60	54.10
SAM-Adapter		SAM-B	55.18	63.10	57.70	67.98	34.55	43.27	54.48	53.75
SAMed		SAM-B	56.03	63.30	58.14	67.33	36.01	43.81	56.09	54.39
RSPrompter		SAM-B	56.02	65.72	56.88	68.88	33.38	44.24	56.20	54.48
Semantic-SAM		SAM-B	55.94	65.61	58.55	68.40	33.19	43.87	56.55	54.59
FE-SAM(Ours)		SAM-B	**56.77**	**66.10**	58.92	69.14	**36.18**	44.35	58.15	**55.66**

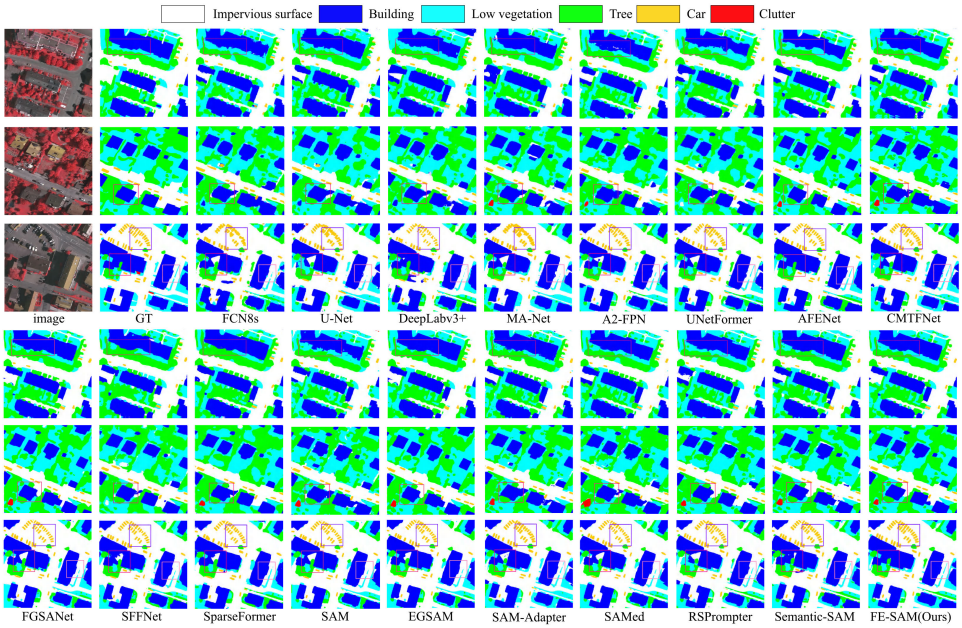


Fig. 6. Additional qualitative results on the ISPRS Vaihingen dataset. (Top to bottom), each row showcases a different challenging scene. Our model, FE-SAM (last column), consistently produces segmentation masks that are more detailed and accurate than those from competing methods.

fragmented road networks where topological connectivity is compromised, and they generate over-smoothed residential areas where individual building footprints are lost or blurred into an indistinct mass. Our FE-SAM, however, exemplifies exceptional adaptability. This is a direct testament to the FMA's capacity to selectively modulate frequency components based on scene content. It can enhance low-frequency components to preserve the coherence of large, continuous structures like roads, while concurrently boosting high-frequency components to capture the intricate details of small, dense building clusters. This scene-specific frequency modulation allows our model to excel at both tasks simultaneously.

3) Evaluation of Boundary Precision: To further evaluate the boundary-sensitive segmentation quality of FE-SAM, we selected several representative models for a comparative study using the B-IoU metric. FE-SAM does not use any boundary-specific loss. Nevertheless, since EGRefiner

TABLE VI

QUANTITATIVE COMPARISON OF BOUNDARY SEGMENTATION QUALITY ON THE ISPRS VAIHINGEN DATASET. B-IOU (d = 8) AND B-IOU (d = 16) MEASURE THE ACCURACY WITHIN 8 AND 16 PIXELS FROM THE BOUNDARY

Method	Backbone	mIoU (%)	B-IoU ($d = 8$)	B-IoU ($d = 16$)
U-Net[42]	ResNet-18	80.96	73.12	75.48
DeepLabv3+[43]	ResNet-18	81.78	74.56	76.82
UNetFormer[46]	ResNet-18	82.86	76.28	78.53
SAM[8]	ViT-B	83.04	75.82	77.96
SAM-Adapter[50]	SAM-B	83.35	77.14	79.21
Semantic-SAM[52]	SAM-B	84.56	79.43	81.67
FE-SAM (Ours)	SAM-B	**85.24**	**82.15**	**84.32**

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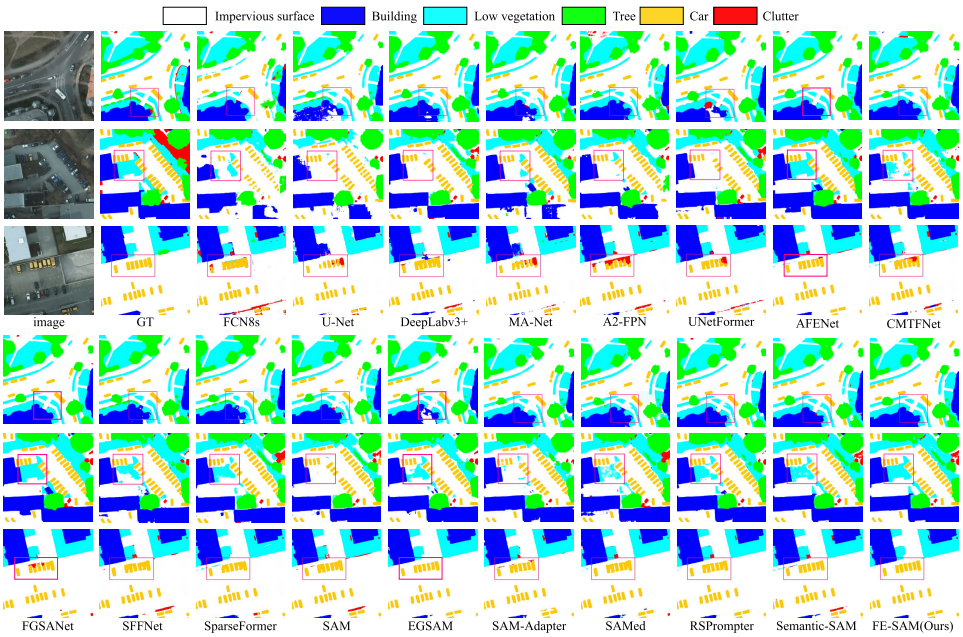


Fig. 7. Additional qualitative results on the ISPRS Potsdam dataset. Our model, FE-SAM (last column), consistently generates more accurate and complete segmentation masks compared to other state-of-the-art methods, especially in areas with complex object layouts.



Fig. 8. Additional qualitative results on the LoveDA dataset. The selected scenes showcase diverse urban and rural landscapes. Our model, FE-SAM (last column), consistently provides more complete and semantically accurate segmentation masks than competing methods.

introduces multiscale edge-enhanced cues from the original image, we report B-IoU as an auxiliary metric to evaluate boundary-sensitive segmentation quality. B-IoU is used only for evaluation and is not involved in training. As presented in Table VI, FE-SAM achieves the highest B-IoU scores across different dilation widths ($d = 8$ and $d = 16$) on the Vaihingen dataset. Compared to the state-of-the-art Semantic-SAM, our model provides a more significant margin of improvement in B-IoU than in global mIoU. For instance, the B-IoU ($d = 8$) of FE-SAM reaches 82.15%, which is 2.72% higher than the second-best method. These results indicate that EGRefiner improves boundary-sensitive segmentation quality, especially for structures such as building corners and narrow roads.

4) Model Complexity Comparison: We analyze the complexity of FE-SAM to assess its practical viability, and the experimental results are shown in Table VII and Fig. 9. Our method introduces a modest overhead of only 7.58 M trainable parameters (mere 8.7% increase over the original SAM), which yields a substantial performance gain (+2.2 mIoU on ISPRS Vaihingen). While this represents a slightly larger parameter budget than simpler methods like SAM-Adapter, the significant performance improvement validates the efficacy of our

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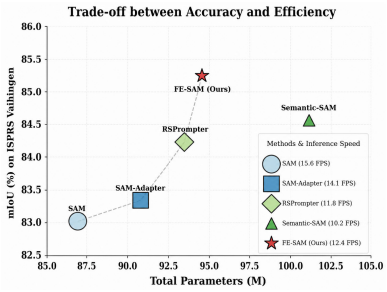


Fig. 9. Comparison of mIoU, total parameters, and inference speed among SAM-based methods.

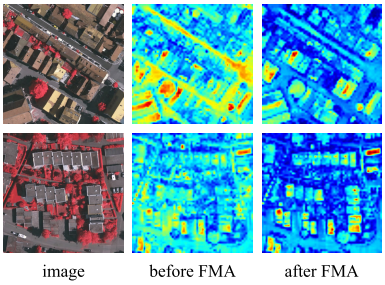


Fig. 10. Visual comparisons of feature representations before and after the FMA module. (Left to right) Original remote sensing images, feature heatmaps before FMA, and feature heatmaps after FMA.

TABLE VII

MODEL COMPLEXITY COMPARISON OF SAM-BASED METHODS

Method	Total Params(M)	FLOPs(M)	FPS
SAM [8]	86.94	103.17	15.6
SAM-Adapter [50]	90.78	109.61	14.1
RSPrompter [18]	93.45	113.66	11.8
Semantic-SAM [52]	101.12	115.83	10.2
FE-SAM (Ours)	94.52	114.11	12.4

Note: The parameters of the frozen prompt encoder are included in all complexity statistics for fairness.

TABLE VIII

ABLATION STUDIES ON DIFFERENT MODULES OF FE-SAM

FMA	EGRefiner	Vaihingen	Potsdam	LoveDA
		83.04	85.94	52.54
✓		84.06	86.50	54.13
	✓	83.57	86.25	53.64
✓	✓	**85.24**	**86.73**	**55.66**

specialized FMA and EGRefiner modules, which are tailored to the physical properties of remote sensing data. The FFT and IFFT operations in FMA are computationally efficient on modern hardware, and the EGRefiner follows a bilateral fusion philosophy [57] to balance spatial refinement and inference speed. FE-SAM achieves a competitive inference speed of 12.4 FPS. The efficiency of our FE-SAM is most evident when compared to the strong competitor, semantic-SAM. FE-SAM not only achieves superior accuracy but does so more efficiently, requiring 6.6 M fewer parameters and 1.7 G fewer FLOPs. It shows that our state-of-the-art performance is a result of an intelligent and targeted architecture rather than simply scaling model size.

C. Ablation Studies

To validate the contributions of different components in our FE-SAM, we conducted a series of ablation studies. This section interrogates the efficacy of our core modules, the architectural principles of the FMA, and the structural integrity of the EGRefiner. All ablations are performed on the challenging ISPRS Vaihingen dataset.

As presented in Table VIII, applying the FMA module alone improves the baseline mIoU from 83.04% to 84.06%, a significant gain confirming that our frequency-domain adaptation is highly effective at aligning SAM's features with remote sensing data. Similarly, integrating the EGRefiner alone boosts performance to 83.57%. When both modules are used in our FE-SAM, the performance reaches 85.24%. It demonstrates that FMA and EGRefiner are not redundant but are highly complementary, working in synergy to address the distinct and orthogonal challenges of feature adaptation (FMA) and boundary refinement (EGRefiner), respectively.

TABLE IX

COMPARISON OF DIFFERENT PEFT STRATEGIES. PERFORMANCE IS MEASURED IN MIOU(%)

Method	Trainable Params (M)	Vaihingen	Potsdam	LoveDA
Full tuning	83.02	83.04	85.94	52.54
Adapter	7.83	83.12	86.09	53.14
LoRA	7.09	83.20	86.15	53.07
FMA (Ours)	**7.56**	**84.06**	**86.50**	**54.13**

To intuitively illustrate how the FMA module facilitates feature adaptation, Fig. 10 visualizes the deep feature maps before and after FMA refinement. For each representative remote sensing patch, we present the original image alongside its corresponding feature heatmaps. The raw feature responses before FMA (middle column) are relatively diffuse, heavily entangled with redundant low-frequency activations from complex backgrounds (e.g., roads and flat terrain). Conversely, after applying FMA (right column), this background noise is significantly suppressed. The refined responses sharply converge onto discriminative object structures and high-frequency geometric boundaries. This visual evidence demonstrates that FMA successfully recalibrates frequency-domain representations, effectively enhancing the adaptability and target-awareness of SAM features in dense remote sensing scenes.

D. Analysis of the Frequency-Modulated Adapter (FMA)

1) Comparison With Different PEFT Strategies: To compare the proposed FMA with other PEFT strategies, we conducted a series of experiments, and the results are listed in Table IX. It can be seen that our FMA consistently achieves the best performance across all three datasets. While LoRA and generic adapters also offer parameter efficiency, our FMA surpasses them with significantly higher mIoU scores using a comparable or even smaller budget of trainable parameters. It demonstrates that the FMA is an efficient PEFT method

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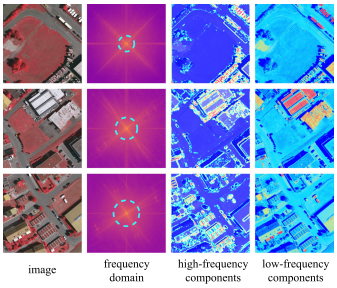


Fig. 11. Visual validation of FMA's adaptive frequency partitioning on the ISPRS Vaihingen dataset.

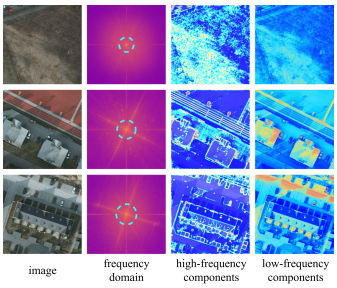


Fig. 12. Visual validation of FMA's adaptive frequency partitioning on the ISPRS Potsdam dataset.

TABLE X

ABLATION ON THE NECESSITY OF FMA'S ADAPTIVE PARTITIONING, EVALUATED ACROSS ALL THREE BENCHMARK DATASETS. PERFORMANCE IS MEASURED IN MIOU(%)

FMA Configuration	Vaihingen	Potsdam	LoveDA
Baseline (w/o FMA)	83.57	86.25	53.64
Fixed LPF/HPF (Generic)	84.12	86.41	54.02
Fixed LPF/HPF (Learned)	84.45	86.50	54.31
FMA (Ours)	**85.24**	**86.73**	**55.66**

for high-resolution remote sensing image segmentation. It achieves a favorable tradeoff between efficiency and accuracy.

2) Necessity of Adaptive Frequency Partitioning: To prove that the effectiveness of adaptive frequency partitioning of FMA, we first quantitatively establish its superiority over conventional, nonadaptive frequency separation techniques in Table X. Our adaptive FMA surpasses both the generic fixed filter and the learned static filter by a considerable margin across all three datasets. It confirms that a static, one-size-fits-all frequency cutoff is insufficient for the diverse characteristics of remote sensing scenes. We further visually validate how our FMA addresses the challenge of adapting SAM's features to diverse land-cover types. As shown in Figs. 11 and 12, the frequency partitioning is dictated by an adaptive radius r (visualized as a dashed circle in the Fourier domain), which is dynamically computed by accumulating spectral energy radially from the low-frequency origin until a predefined energy percentage p is satisfied. As shown in Fig. 11, when the image is full of intricate edges from buildings and cars, the FMA is compelled to integrate over a wider spectral area—systematically computing a larger radius r—a phenomenon consistently visible across these urban examples.

TABLE XI

ABLATION STUDY ON THE MULTISCALE ARCHITECTURE OF EGREFINER ON THREE BENCHMARK DATASETS (MIOU%)

Configuration	Vaihingen	Potsdam	LoveDA
Single-Scale DConv 3x3	84.72	86.55	54.71
Single-Scale DConv 5x5	84.85	86.60	55.06
Single-Scale DConv 7x7	84.81	86.58	54.89
Multi-Scale DConv (Ours)	**85.24**	**86.73**	**55.66**

TABLE XII

ABLATION STUDY ON THE ARCHITECTURAL COMPONENTS OF THE EGREFINER, EVALUATED ACROSS ALL THREE BENCHMARK DATASETS

EGRefiner Configuration	Vaihingen	Potsdam	LoveDA
w/o Element-wise Subtraction	84.65	86.48	55.03
w/o Channel-wise Concat	84.81	86.55	55.21
w/o F_s Feature Fusion	79.32	81.05	49.58
EGRefiner (Full, Ours)	**85.24**	**86.73**	**55.66**

E. Analysis of the EGRefiner

We conduct extensive experiments to verify the effectiveness of EGRefiner. The multiscale feature extraction and different components are analyzed in detail.

1) Effectiveness of Multiscale Feature Extraction: The design of EGRefiner is predicated on the hypothesis that a comprehensive edge representation for high-resolution remote sensing imagery requires fusing information from multiple receptive fields. We first validated this by comparing our multiscale dilated convolution design against single-scale configurations. As presented in Table XI, the multiscale DConv (Ours) approach achieves a superior mIoU of 85.24% on the ISPRS Vaihingen dataset, outperforming all single-scale variants. This confirms that the synergistic aggregation of features is crucial for simultaneously capturing both fine-grained details via small kernels and larger structural information via larger kernels.

2) Analysis of Critical Components in EGRefiner: We conducted an ablation study to validate the contribution of the EGRefiner's key components (see Table XII). Removing the elementwise subtraction or replacing channelwise concatenation with summation leads to a consistent and significant performance drop, confirming their importance in the EGRefiner. Furthermore, removing the fusion with the decoder's semantic features (F_s) causes a catastrophic drop in performance, with the mIoU falling far below even the baseline model without an EGRefiner. It is evident that low-level edge information is counterproductive when used in isolation. Without the high-level semantic context provided by F_s, the model is unable to distinguish between meaningful object boundaries and irrelevant high-frequency texture noise.

3) Qualitative Analysis of Boundary Refinement: We present visual comparisons in Fig. 13 on challenging scenes from the ISPRS Vaihingen dataset. The results offer compelling visual evidence of the EGRefiner's critical contribution. Existing methods produce segmentation results with notably blurred edges. For instance, in the top row, the sharp corners of the building are rendered as smooth curves by the other methods. Similarly, in the bottom row, the other

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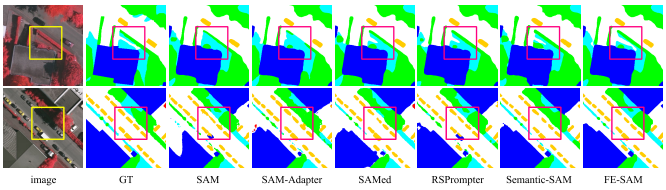


Fig. 13. Qualitative comparison visualizing the efficacy of the EGRefiner for boundary refinement on challenging scenes from the ISPRS Vaihingen dataset.

methods do not perform well in car segmentation. On the contrary, our FE-SAM, empowered by the EGRefiner, delivers superior segmentation results. The visual comparisons validate that the EGRefiner has superior segmentation performance in fine-grained remote sensing applications.

V. Conclusion

In this article, we proposed FE-SAM, a frequency- and edge-guided framework for adapting the SAM to RSISS. To alleviate the domain gap between natural and remote sensing images, the FMA performs input-adaptive frequency separation and prototype-based recalibration, enabling SAM to selectively enhance informative high- and low-frequency components with only a small number of trainable parameters. Meanwhile, the EGRefiner introduces multiscale structural cues from the original image to recover spatial details and improve object boundary delineation. Extensive experiments on the ISPRS Vaihingen, ISPRS Potsdam, and LoveDA datasets demonstrate that FE-SAM consistently outperforms state-of-the-art methods.

The proposed frequency adaptation and edge-guided refinement strategies are also applicable to other dense prediction tasks, including medical image segmentation, aerial scene parsing, and change detection. Recent studies, such as FireViT [58], have validated the effectiveness of vision transformer architectures in complex visual understanding. Inspired by these advances, FE-SAM offers a general feature adaptation framework that can be extended to diverse dense prediction scenarios demanding adaptive feature learning and precise boundary localization.

References

[1] D. Zhao, J. Lu, and B. Yuan, "See, perceive, and answer: A unified benchmark for high-resolution postdisaster evaluation in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5618914.

[2] X. Zhang et al., "Remote sensing image semantic change detection boosted by semi-supervised contrastive learning of semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5624113.

[3] C. Ye, Y. Zhuge, and P. Zhang, "Towards open-vocabulary remote sensing image semantic segmentation," in *Proc. AAAI Conf. Artif. Intell.*, 2025, pp. 9436–9444.

[4] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504218.

[5] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117.

[6] Y. Hua, D. Marcos, L. Mou, X. X. Zhu, and D. Tuia, "Semantic segmentation of remote sensing images with sparse annotations," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[7] R. Liu, L. Mi, and Z. Chen, "AFNet: Adaptive fusion network for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7871–7886, Sep. 2021.

[8] X. Ma, X. Zhang, M.-O. Pun, and M. Liu, "A multilevel multimodal fusion transformer for remote sensing semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5403215.

[9] X. Li et al., "AAFormer: Attention-attended transformer for semantic segmentation of remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[10] A. Kirillov et al., "Segment anything," in *Proc. Int. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 3992–4003.

[11] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021, pp. 1–13.

[12] X. Yan et al., "AFTer-SAM: Adapting SAM with axial fusion transformer for medical imaging segmentation," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 7960–7969.

[13] W. Xie, N. Willems, S. Patil, Y. Li, and M. Kumar, "SAM fewshot finetuning for anatomical segmentation in medical images," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 3241–3249.

[14] X. Wei, J. Cao, Y. Jin, M. Lu, G. Wang, and S. Zhang, "I-MedSAM: Implicit medical image segmentation with segment anything," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 90–107.

[15] Z. Yu, X. Zhang, L. Zhao, Y. Bin, and G. Xiao, "Exploring deeper! Segment anything model with depth perception for camouflaged object detection," in *Proc. 32nd ACM Int. Conf. Multimedia*, Oct. 2024, pp. 4322–4330.

[16] M. N. Meeran, G. Adethya T, and B. P. Mantha, "SAM-PM: Enhancing video camouflaged object detection using spatio-temporal attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2024, pp. 1857–1866.

[17] W. Hui, Z. Zhu, S. Zheng, and Y. Zhao, "Endow SAM with keen eyes: Temporal-spatial prompt learning for video camouflaged object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 19058–19067.

[18] D. Zhang, F. Wang, L. Ning, Z. Zhao, J. Gao, and X. Li, "Integrating SAM with feature interaction for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4513011.

[19] Z. Shan, Y. Liu, L. Zhou, C. Yan, H. Wang, and X. Xie, "ROSSAM: High-quality interactive segmentation for remote sensing moving object," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 3625–3635.

[20] K. Chen et al., "RSPrompter: Learning to prompt for remote sensing instance segmentation based on visual foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4701117.

[21] Z. Yan et al., "RingMo-SAM: A foundation model for segment anything in multimodal remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5625716.

[22] W. Sun and R. Wang, "Fully convolutional networks for semantic segmentation of very high resolution remotely sensed images combined with DSM," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 3, pp. 474–478, Mar. 2018.

[23] B. Cheng, C. Liang, X. Liu, Y. Liu, X. Ma, and G. Wang, "Research on a novel extraction method using deep learning based on GF-2 images for aquaculture areas," *Int. J. Remote Sens.*, vol. 41, no. 9, pp. 3575–3591, May 2020.

[24] R. Li, S. Zheng, C. Zhang, C. Duan, L. Wang, and P. M. Atkinson, "ABCNet: Attentive bilateral contextual network for efficient semantic segmentation of fine-resolution remotely sensed imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 181, pp. 84–98, Nov. 2021.

[25] M. Yao, Y. Zhang, G. Liu, and D. Pang, "SSNet: A novel transformer and CNN hybrid network for remote sensing semantic segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 3023–3037, 2024.

[26] H. Wu, M. Zhang, P. Huang, and W. Tang, "CMLFormer: CNN and multiscale local-context transformer network for remote sensing images semantic segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 7233–7241, 2024.

[27] X. Xiang, W. Gong, S. Li, J. Chen, and T. Ren, "TCNet: Multiscale fusion of transformer and CNN for semantic segmentation of remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 3123–3136, 2024.

[28] L. Zhang, "Frequency-guided adaptive feature fusion for remote sensing image segmentation," *IEEE Trans. Multimedia*, vol. 27, pp. 72–83, 2025.

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[29] F. Gao, M. Fu, J. Cao, J. Dong, and Q. Du, "Adaptive frequency enhancement network for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5619415.

[30] J. Wu, Y. Jiang, Q. Liu, Z. Yuan, X. Bai, and S. Bai, "General object foundation model for images and videos at scale," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 3783–3795.

[31] M. Awais et al., "Foundation models defining a new era in vision: A survey and putlook," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 4, pp. 2245–2264, Apr. 2025.

[32] X. Sun et al., "RingMo: A remote sensing foundation model with masked image modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5612822.

[33] Y. Cong et al., "SatMAE: Pre-training transformers for temporal and multi-spectral satellite imagery," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 1–13.

[34] M. Noman, M. Naseer, H. Cholakkal, R. M. Anwar, S. Khan, and F. S. Khan, "Rethinking transformers pre-training for multi-spectral satellite imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27811–27819.

[35] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[36] M. T. Usman, H. Khan, I. Rida, and J. Koo, "Lightweight transformer-driven multi-scale trapezoidal attention network for saliency detection," *Eng. Appl. Artif. Intell.*, vol. 155, Sep. 2025, Art. no. 110917.

[37] L. Chen, X. Wang, F. Yao, M. Song, J. Zhang, and S. Wang, "An edge-guided SAM for effective complex object segmentation," *Expert Syst. Appl.*, vol. 271, May 2025, Art. no. 126560.

[38] S. Ravishankar and G. Battineni, "A survey on recent advancements in auto-machine learning with a focus on feature engineering," *J. Comput. Cognit. Eng.*, vol. 4, no. 1, pp. 56–63, May 2023.

[39] A. Le, D. A. Pham, D. Thanh, and H. B. Vo, "AlertTrap: A study on object detection in remote insect trap monitoring system using on the edge deep learning platform," *J. Comput. Cognit. Eng.*, vol. 4, no. 3, pp. 284–295, Jun. 2024.

[40] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, "A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[41] L. Mou, Y. Hua, and X. X. Zhu, "Relation matters: Relational context-aware fully convolutional network for semantic segmentation of high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7557–7569, Nov. 2020.

[42] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," in *Proc. Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2021, pp. 1–12.

[43] J. Song, H. Chen, and N. Yokoya, "SyntheWorld: A large-scale synthetic dataset for land cover mapping and building change detection," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 8272–8281.

[44] B. Cheng, R. Girshick, P. Dollár, A. C. Berg, and A. Kirillov, "Boundary IoU: Improving object-centric image segmentation evaluation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15329–15337.

[45] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[46] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351. Cham, Switzerland: Springer, 2015, pp. 234–241.

[47] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with Atrous separable convolution for semantic image segmentation," in *Proc. 15th Eur. Conf. Comput. Vis. (ECCV)*, Munich, Germany, 2018, pp. 833–851.

[48] R. Li et al., "Multiattention network for semantic segmentation of fine-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607713.

[49] R. Li, L. Wang, C. Zhang, C. Duan, and S. Zheng, "A^2-FPN for semantic segmentation of fine-resolution remotely sensed images," *Int. J. Remote Sens.*, vol. 43, no. 3, pp. 1131–1155, Feb. 2022.

[50] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.

[51] H. Wu, P. Huang, M. Zhang, W. Tang, and X. Yu, "CMTFNet: CNN and multiscale transformer fusion network for remote-sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2004612.

[52] Y. Yang, G. Yuan, and J. Li, "SFFNet: A wavelet-based spatial and frequency domain fusion network for remote sensing segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 3000617.

[53] Y. Chen et al., "SparseFormer: A credible dual-CNN expert-guided transformer for remote sensing image segmentation with sparse point annotation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5607316.

[54] T. Chen et al., "SAM-adapter: Adapting segment anything in underperformed scenes," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2023, pp. 3367–3375.

[55] Y. Zhang, Z. Shen, and R. Jiao, "Segment anything model for medical image segmentation: Current applications and future directions," *Comput. Biol. Med.*, vol. 171, pp. 1–14, Mar. 2024.

[56] F. Li et al., "Segment and recognize anything at any granularity," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024, pp. 467–484.

[57] H. Khan, M. T. Usman, and J. Koo, "Bilateral feature fusion with hexagonal attention for robust saliency detection under uncertain environments," *Inf. Fusion*, vol. 121, Sep. 2025, Art. no. 103165.

[58] X. Liu and V. Y. Mariano, "The fire-ViT model for tunnel fire detection with vision transformer improvement," *J. Comput. Cognit. Eng.*, vol. 4, no. 1, pp. 89–96, Sep. 2024.

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